PCA (principle component analysis) is one of the oldest method for linear dimensionality reduction.

Two views of PCA: find a lower dimension subspace such that

1. keep the most sample variance

2. minimize reconstruction loss.

Data: $X \in \mathbb{R}^{n \times m}$, and we want to project X into $\mathbb{R}^d$ where $d < m$.

↳ assume it is centered (after substracting its mean for each dimension)

Then projection matrix $U \in \mathbb{R}^{m \times d}$ where $u_i^T u_j = \begin{cases} 1 & \text{if } i=j \\ 0 & \text{otherwise} \end{cases}$ (note this implies $\|u_i\| = 1$)

1. keep the most sample variance after projection

$$\arg\max_U \text{trace}(U^T X^T X U)$$

2. minimize the reconstruction error

$$\arg\min_U \| X^T - UU^T X^T \|_2^2$$

$UU^T X^T$: project; U: recover

Let us try to solve $\arg\max_U \text{trace}(U^T X^T X U)$ Subject $u_i^T u_j = \begin{cases} 1 & \text{if } i=j \\ 0 & \text{otherwise} \end{cases}$

$\Downarrow$

$U^T U = I$

Lagrangian form

$$\mathcal{L}(U, \Lambda) = \text{trace}(U^T X^T X U) - \text{trace}(\Lambda(U^T U - I))$$

each constraint will done on λ. Λ must be symmetric, since $u_i^T u_j = u_j^T u_i$

constraints

$\text{trace}(\vec{A}\vec{B}) = \sum_{i,j} A_{i,j} B_{i,j}$ (sums the elementwise products of A and B)

Then $\frac{\partial \mathcal{L}}{\partial U} = 2X^T X U - 2U\Lambda = 0$

$\Rightarrow X^T X U = U\Lambda$ → force it to be diagonal, then U must be eigenvector of $X^T X$ and Λ must be eigenvalues, to maximize $\text{trace}(U^T X^T X U) = \text{trace}(\Lambda)$. We choose the maximum eigenvalues

$\frac{\partial \mathcal{L}}{\partial \Lambda} = U^T U - I = 0 \Rightarrow U^T U = I$

Why can we force Λ to be diagonal? We show that solution W with corresponding $\Lambda = \Lambda'$, we can find another solution W' corresponding to a diagonal Λ.

Suppose the optimal solution for U is W with corresponding $\Lambda = \Lambda'$. Since Λ' is symmetric, we must have

$$\Lambda' = Q D Q^T$$

We first show that $W' = WQ$ is also an optimal solution

1. W' satisfies the constraints

$$\begin{aligned} W'^T W' &= Q^T W^T W Q \\ &= Q^T Q \\ &= I \end{aligned}$$

2. W' leads to the same solution as W

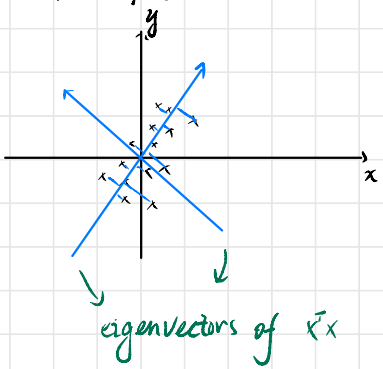
$$\begin{aligned} \text{trace}(W'^T X^T X W') &= \text{trace}(Q^T W^T X^T X W Q) \\ &= \text{trace}(QQ^T W^T X^T X W) \\ &= \text{trace}(W^T X^T X W) \end{aligned}$$

Then, we show that W' corresponding to a symmetric Λ.

Since W' is an optimal solution, we must have

$$\begin{aligned} W'^T X^T X W' &= \Lambda \\ Q^T W^T X^T X W Q &= \Lambda \\ Q^T \Lambda' Q &= \Lambda \\ Q^T Q D Q^T Q &= \Lambda \\ D &= \Lambda \end{aligned}$$

In short, PCA projects the data into the first d eigenvectors of X^TX.



A few notes about PCA:

1. in practice, PCA will first standarize the data for each dimension. If we don't center the data, the direction of the eigenvector will be biased. If we don't scale each dimension, then the dimension with larger variance will be favoured, i.e. get more variance after projection.

2. after projection and reconstruction the data is UU^TX^T with covarian matrix:

$$(UU^TX^T)(XUU^T) = UU^TX^TXUU^T$$
$$= UU^TU\Lambda U^T \quad \left(U\Lambda = U\begin{pmatrix}\lambda_1 & & \\ & \ddots & \\ & & \lambda_d\end{pmatrix}\right)$$
$$= U\Lambda U^T$$

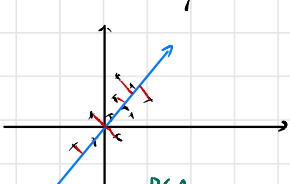
$U_{:,i}$ is eigenvector of X^TX

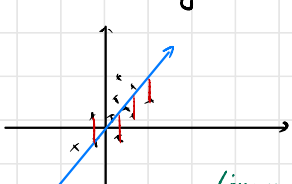
note that: $\text{trace}(U\Lambda U^T) = \text{trace}(\Lambda)$, i.e. the variance equals the sum of eigenvalues

3. if we only do projection (without reconstruction), the data is

U^TX^T, with covariance matrix Λ (uncorrelated)

4. notice how PCA is different than linear regression.


PCA


linear regression

PCA is quite intuitive, but how do we evaluate the quality of PCA?

1. Check the variance kept

2. generative approach, i.e. probabilistic PCA (PPCA), where we seek Gaussian latent variable model. It will allow likelihood evaluation, outlier detection and data generation.

PPCA: Suppose we have data $X \in \mathbb{R}^{n \times m}$ (n: sample size), and as before, we want to find a d-dimensional linear space to project the data into.

1. Assume each data point x_i has an unknown $z_i \in \mathbb{R}^d$ associate with it (in the subspace). This variable z_i such that

$$z_i \in \mathcal{N}(0, I_d)$$

2. Each x_i is the outcome of a random variable such that

$$x_i | z_i \sim \mathcal{N}(Wz_i + \mu, \sigma^2 I_m)$$

In short, we have $x = Wz + \mu + \epsilon$, where

$$z \sim \mathcal{N}(0, I_d)$$

$$\epsilon \sim \mathcal{N}(0, \sigma^2 I_m)$$

$$\begin{aligned} E(x) &= E(Wz + \mu + \epsilon) \\ &= E(Wz) + \mu + E(\epsilon) \\ &= \mu \end{aligned}$$

$$\begin{aligned} Cov(x) &= E\left((Wz + \mu + \epsilon - E(x))(Wz + \mu + \epsilon - E(x))^T \right) \\ &= E\left((Wz + \epsilon)(z^T W^T + \epsilon^T) \right) \\ &= E\left(Wzz^T W^T + \epsilon\epsilon^T \right) \\ &= W E(zz^T) W^T + E(\epsilon\epsilon^T) \\ &= WW^T + \sigma^2 I_m \end{aligned}$$

$$x \sim \mathcal{N}(\mu, WW^T + \sigma^2 I_m)$$

Learn μ, W and σ through MLE

$$\Rightarrow \mu = \bar{x}$$

$$\Rightarrow W = Q(\Lambda - \sigma^2 I)^{\frac{1}{2}} R$$

- Q: eigenvectors
- $\Lambda = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_d \end{pmatrix}$
- R: any orthogonal matrix (we let it = 1 wlog)

$$\Rightarrow \sigma^2 = \frac{1}{m-d} \sum_{i=d+1}^{m} \lambda_i$$

Note that

$$\begin{aligned}\mathrm{Cov}(X, Z) &= E\left((X - E(X))(Z - E(Z))^T\right) \\ &= E\left((X-\mu)Z^T\right) \\ &= E\left(XZ^T\right) \\ &= E\left(WZZ^T\right) \\ &= W\end{aligned}$$

So we have $\begin{pmatrix} X \\ Z \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu \\ 0 \end{pmatrix}, \begin{pmatrix} WW^T + \sigma^2 I_m & W \\ W & I_d \end{pmatrix}\right)$

$\Rightarrow Z|X \sim \mathcal{N}(M^{-1}W^T(X-\mu), \sigma^2 M^{-1})$, where $M = W^TW + \sigma^2 I$

1. Connection with PCA. When $\sigma \to 0$, we have

$$\begin{aligned}E(Z|X) &= (W^TW)^{-1}W^T(X-\mu) \\ &= \left(\Lambda^{\frac{1}{2}}Q^TQ\Lambda^{\frac{1}{2}}\right)^{-1}W^T(X-\mu) \\ &= \Lambda^{-1}\Lambda^{\frac{1}{2}}Q^T(X-\mu) \\ &= \Lambda^{-\frac{1}{2}}Q^T(X-\mu)\end{aligned}$$

$$\begin{aligned}\Rightarrow WE(Z|X) &= Q\Lambda^{\frac{1}{2}}\Lambda^{-\frac{1}{2}}Q^T(X-\mu) \\ &= QQ^T(X-\mu)\end{aligned}$$ equal to PCA

2. But for $\sigma > 0$, we have

$W_{MLE} = Q(\Lambda - \sigma^2 I)^{\frac{1}{2}}$

$$\begin{aligned}WE(Z|X) &= Q(\Lambda-\sigma^2 I)^{\frac{1}{2}}\Lambda^{-1}(\Lambda-\sigma^2 I)^{\frac{1}{2}}Q^T(X-\mu) \\ &= Q(\Lambda-\sigma^2 I)^{\frac{1}{2}}\Lambda^{-\frac{1}{2}}(\Lambda-\sigma^2 I)^{\frac{1}{2}}Q^T(X-\mu) \\ &= Q(I-\sigma^2\Lambda^{-1})^{\frac{1}{2}}(I-\sigma^2\Lambda^{-1})^{\frac{1}{2}}Q^T(X-\mu) \\ &= Q(I-\sigma^2\Lambda^{-1})Q^T(X-\mu)\end{aligned}$$

$1 - \frac{\sigma^2}{\lambda_i}$: Scaling factor before reconstruction